

Parallel Shape-Objective Follow-Up Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `parallel_shape_objective_followup_2026_07_21` pilot campaign. The campaign tested three `polished_dataset` `setpoint` `Fw` arms after the prior shape-gate v2 branch failed bounded promotion:

- a windowed `periodic_gru_sequence` continuation with the stronger shape objective;
- a non-windowed `periodic_mlp_harmonic` arm;
- a curve-aware residual arm.

All three remote runs completed. The scalar campaign leader is

`te_shape_objective_periodic_mlp_harmonic_fw_polished_setpoints` with test MAE `0.001236`. This is a pilot scalar result, not an official promotion. Promotion still requires the bounded `TE Curve Verification Pipeline` shape-first screen against both windowed and non-windowed references.

Campaign Artifacts

Artifact	Path
Campaign output	<code>output/training_campaigns/2026-07-21-18-52-44_parallel_shape_objective_followup_2026_07_21</code>
Campaign leaderboard	<code>output/training_campaigns/2026-07-21-18-52-44_parallel_shape_objective_followup_2026_07_21/campaign_leaderboard.yaml</code>
Campaign best run	<code>output/training_campaigns/2026-07-21-18-52-44_parallel_shape_objective_followup_2026_07_21/campaign_best_run.yaml</code>
Campaign execution report	<code>output/training_campaigns/2026-07-21-18-52-44_parallel_shape_objective_followup_2026_07_21/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/cross_wave/shape_objective/2026-07-21-18-36-30_parallel_shape_objective_followup_campaign_plan_report.md</code>
Technical document	<code>doc/technical/2026-07/2026-07-21/2026-07-21-18-36-30_parallel_shape_objective_followup.md</code>
Track 2 pilot plot config	<code>config/paper_reimplementation/rcim_ml_compensation/reference_family_vs_feedforward/shape_objective_followup_track2_plot_polished_setpoints_fw_matrix.yaml</code>
Track 2 pilot curve plot summary	<code>doc/reports/campaign_results/track_2/verification_plots/shape_objective_followup_polished_setpoints_fw/track2_candidate_curve_plot_summary.yaml</code>
Secondary scalar graph bundle	<code>doc/reports/campaign_results/cross_wave/shape_objective/assets/2026-07-21_parallel_shape_objective_followup</code>

Execution Summary

Field	Value
Campaign name	parallel_shape_objective_followup_2026_07_21
Started at	2026-07-21T18:52:44
Finished at	2026-07-21T19:20:11
Completed runs	3
Failed runs	0
Dataset	polished_dataset
Input mode	setpoints
Surface	Fw
Remote sync note	Manual sync recovery was required after the local SSH wrapper became stale post-training.

Campaign Winner

Field	Value
Run name	te_shape_objective_periodic_mlp_harmonic_fw_polished_setpoints
Run instance	2026-07-21-19-02-58_te_shape_objective_periodic_mlp_harmonic_fw_polished_setpoints
Model family	shape_objective_periodic_mlp_harmonic_fw
Model type	periodic_mlp
Runtime contract	non-windowed pointwise setpoint model
Trainable parameters	28,545
Validation MAE	0.001429
Validation RMSE	0.001867
Test MAE	0.001236
Test RMSE	0.001672

The winning checkpoint is stored at `output/training_runs/shape_objective_followup/2026-07-21-19-02-58_te_shape_objective_periodic_mlp_harmonic_fw_polished_setpoints/checkpoints/periodic_mlp-epoch=025-val_mae=0.00142880.ckpt` .

Pilot Comparison

Family	Surface	Validation MAE	Test MAE	Decision
Non-windowed MLP	Fw	0.001429	0.001236	Pilot scalar leader; requires curve-first screen
Windowed GRU	Fw	0.001820	0.001400	Do not promote from scalar closeout
Curve-aware residual	Fw	0.001972	0.001463	Do not promote from scalar closeout

The non-windowed MLP arm has scalar test MAE 11.7% lower than the windowed GRU continuation and 15.5% lower than the curve-aware residual arm. This answers the branch question directly: the best result in this pilot is not the time-windowed GRU branch.

Metric Breakdown

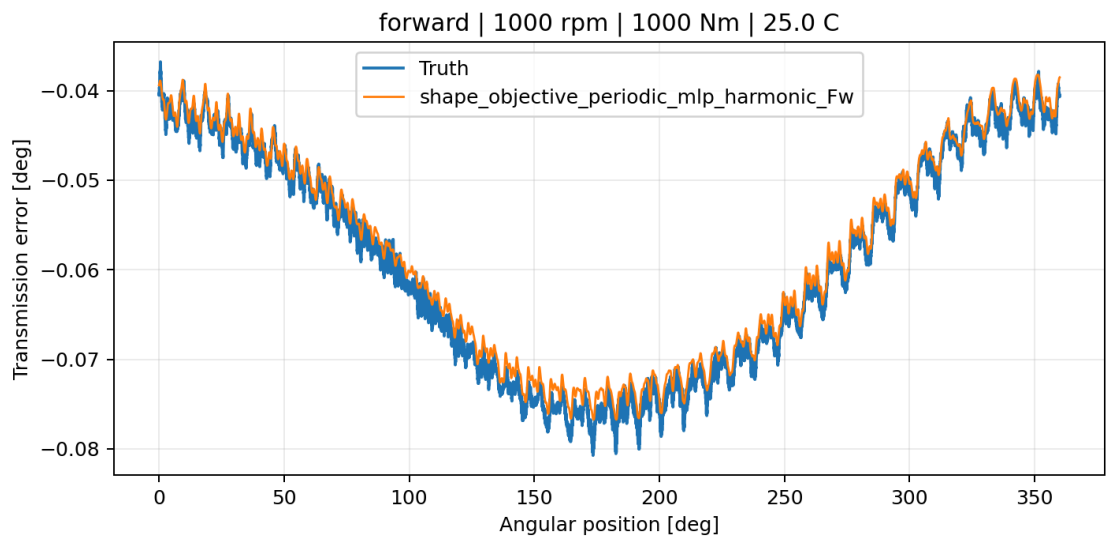
Metric	Validation	Test
MAE	0.001429	0.001236
RMSE	0.001867	0.001672
Centered curve shape loss	0.007328	0.005083
Curve offset loss	0.002797	0.002295
Curve amplitude loss	0.099152	0.079421
Sparse harmonic shape loss	0.000127	0.000086

Pilot Graphs

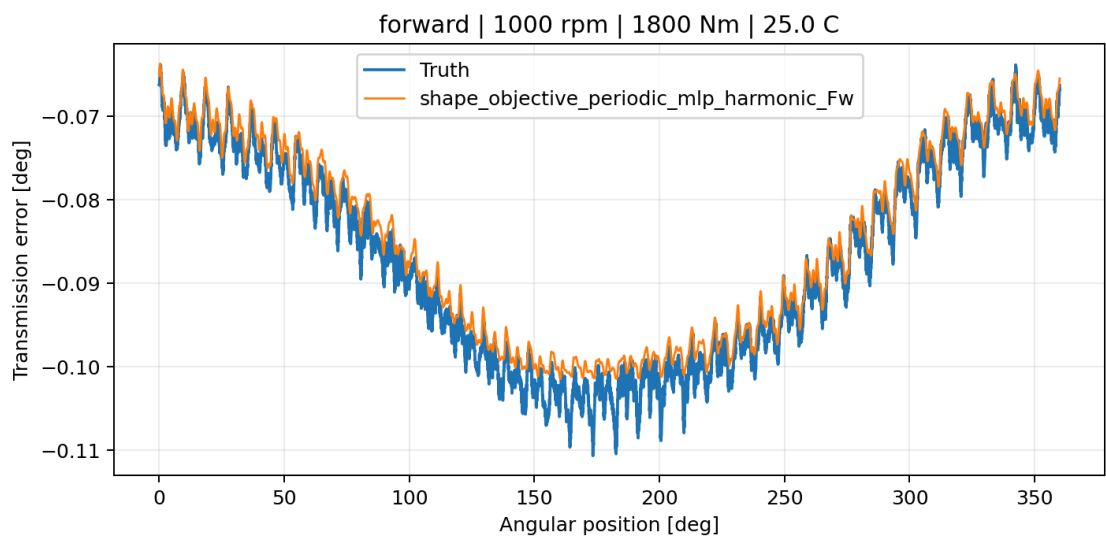
Track 2 Measured-Versus-Predicted Curves

The following plots are bounded Track 2 TE curve overlays. The dark curve is measured TE and the colored curve is the candidate prediction, rendered on held-out `polished_dataset` setpoint `Fw` curves.

shape_objective_periodic_mlp_harmonic_Fw

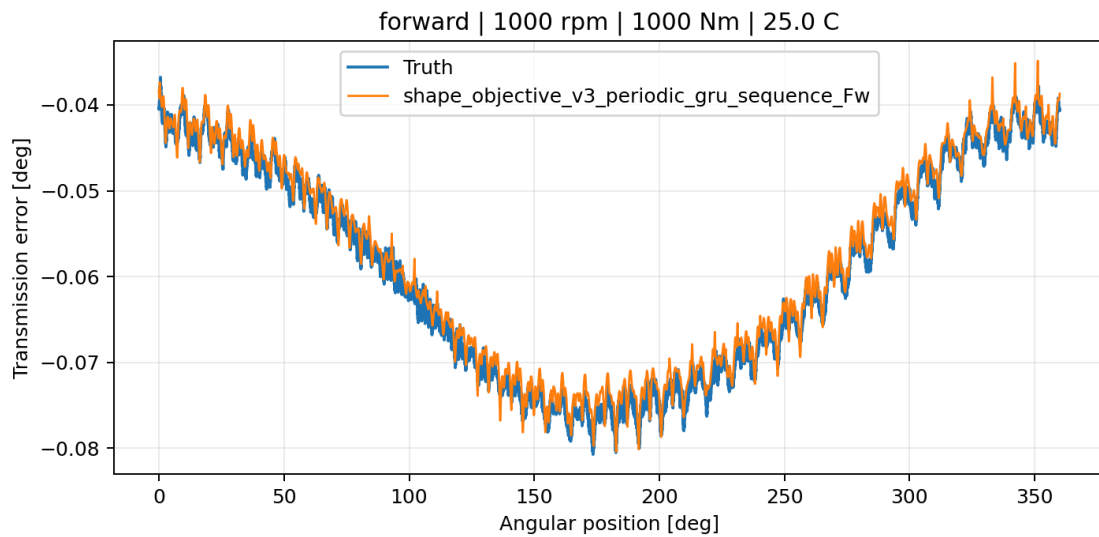


shape_objective_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

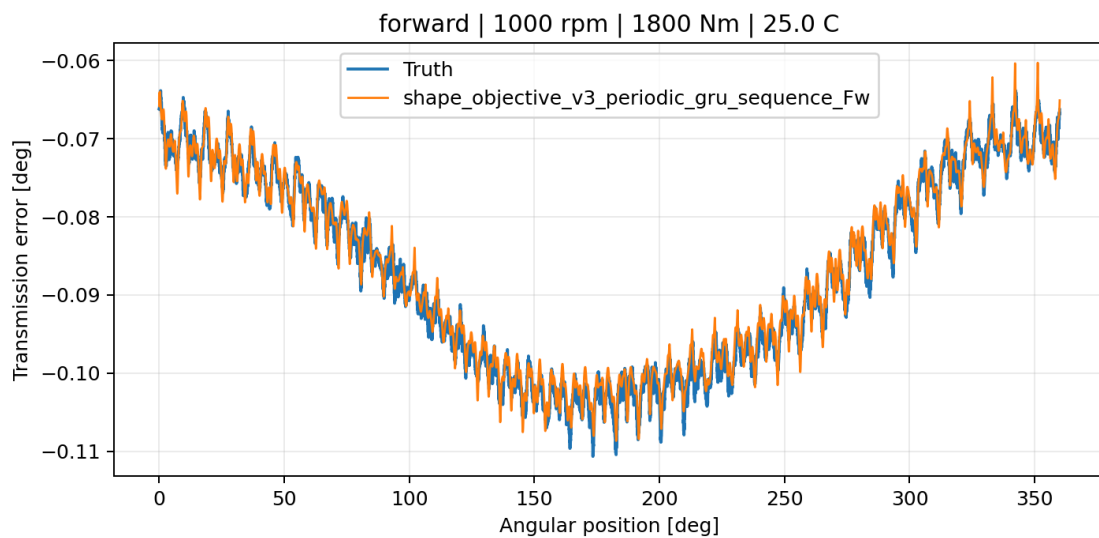


shape_objective_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

shape_objective_v3_periodic_gru_sequence_Fw

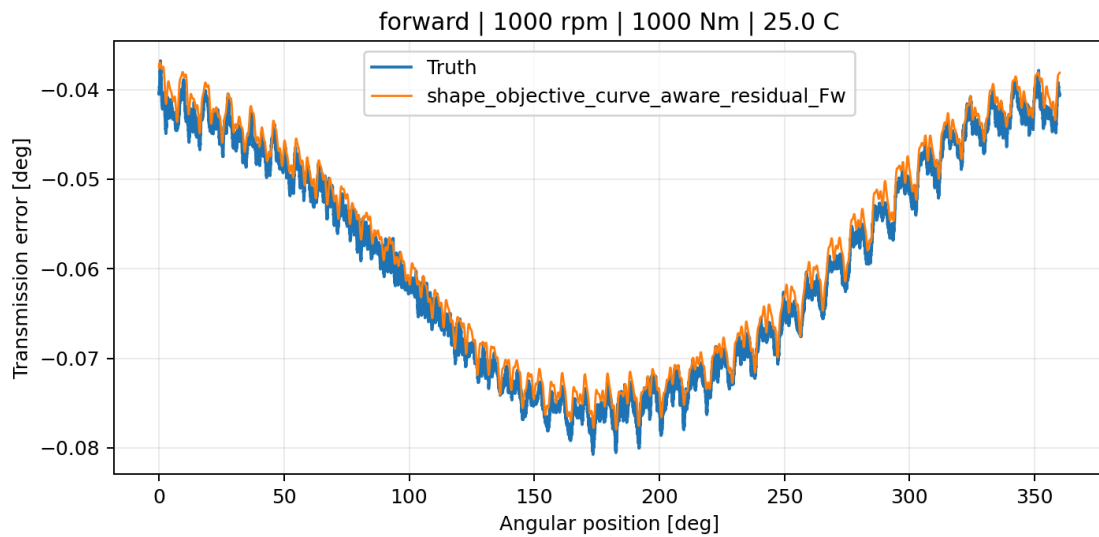


shape_objective_v3_periodic_gru_sequence_Fw measured-versus-predicted TE curve

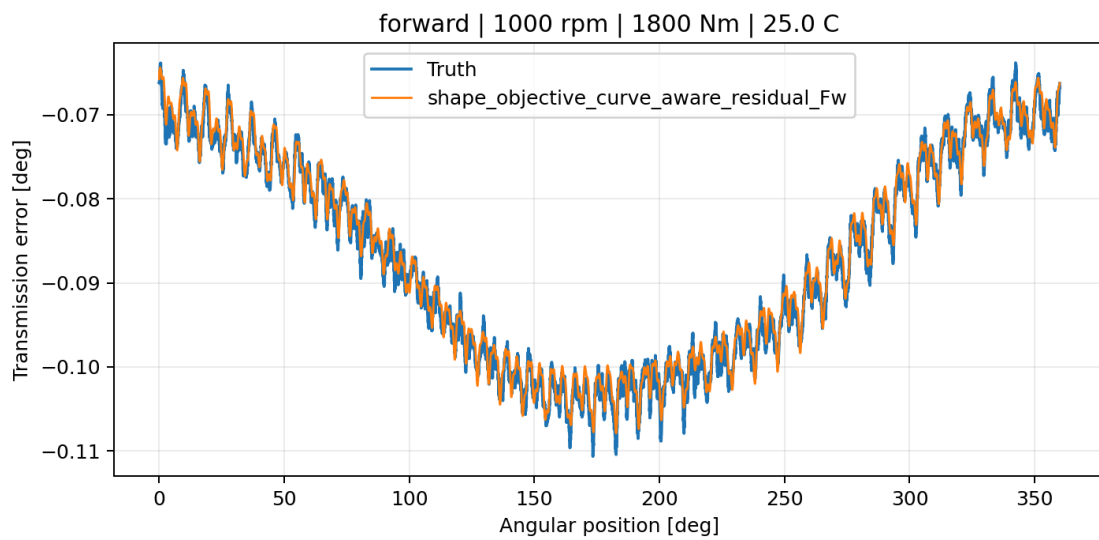


shape_objective_v3_periodic_gru_sequence_Fw measured-versus-predicted TE curve

shape_objective_curve_aware_residual_Fw

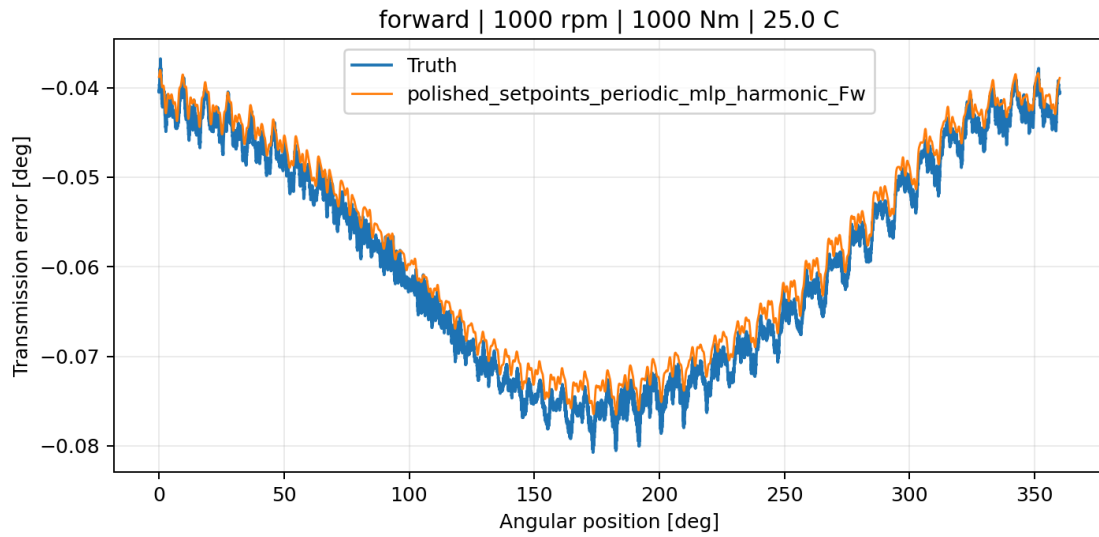


shape_objective_curve_aware_residual_Fw measured-versus-predicted TE curve

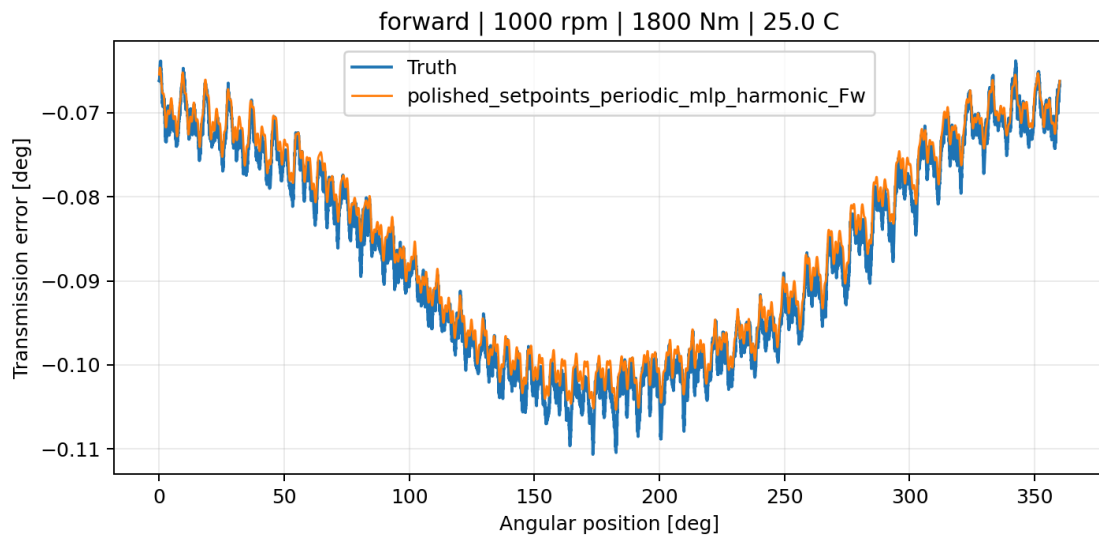


shape_objective_curve_aware_residual_Fw measured-versus-predicted TE curve

polished_setpoints_periodic_mlp_harmonic_Fw

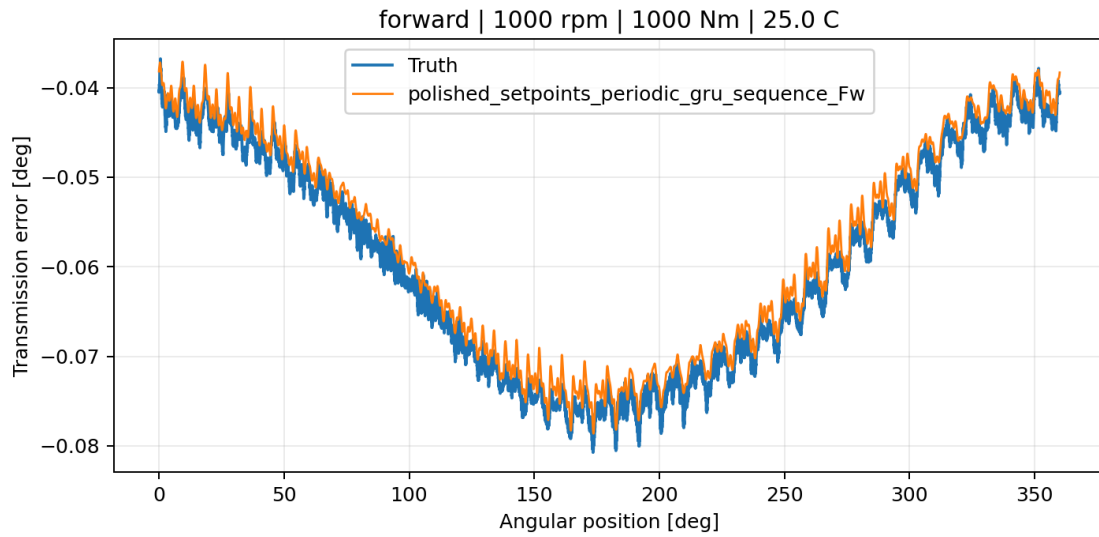


polished_setpoints_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

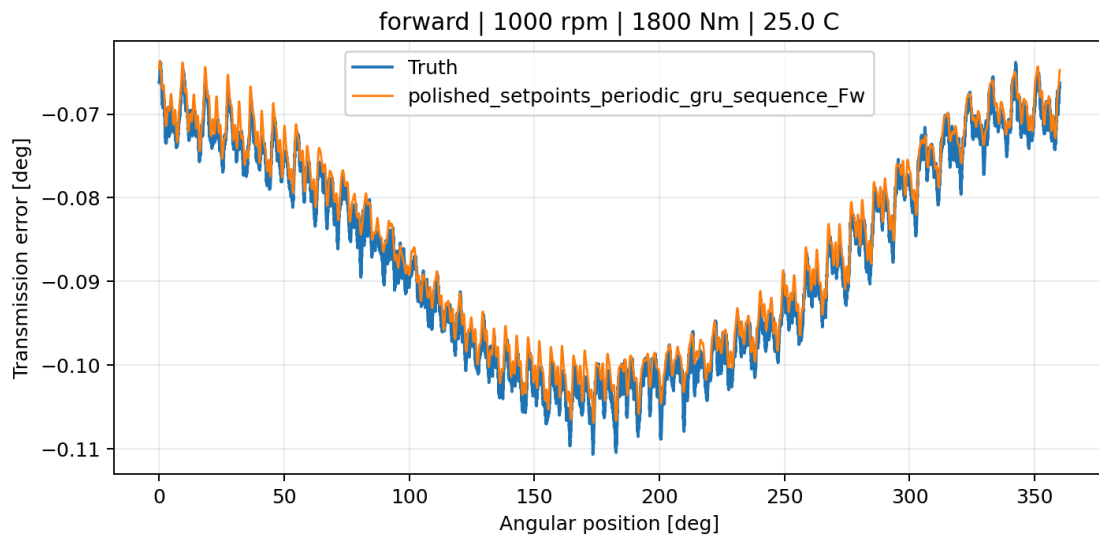


polished_setpoints_periodic_mlp_harmonic_Fw measured-versus-predicted TE curve

polished_setpoints_periodic_gru_sequence_Fw

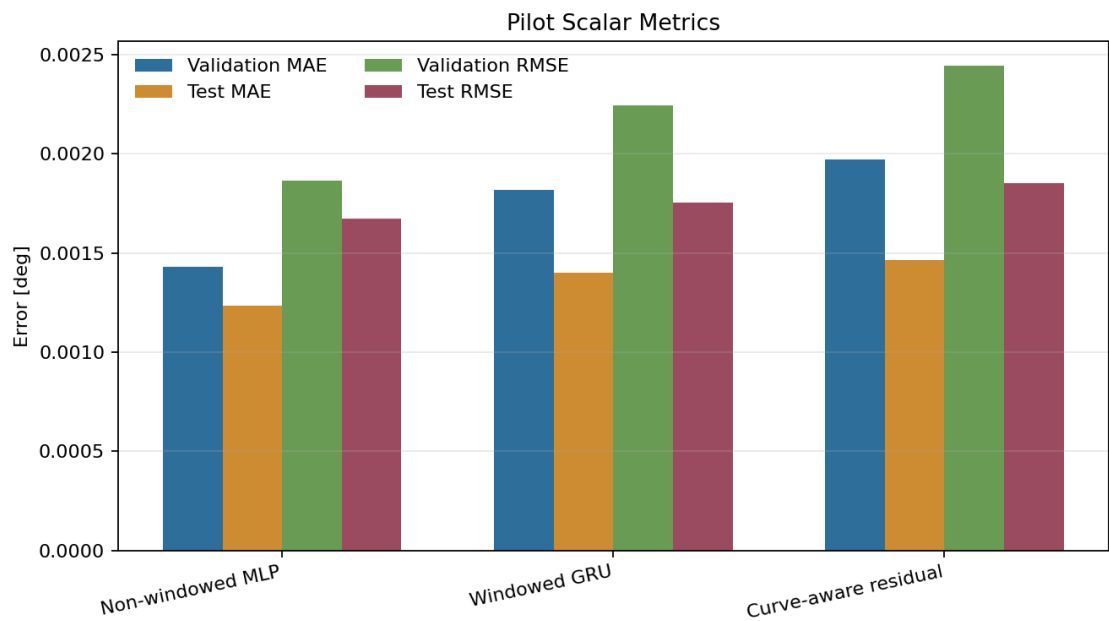


polished_setpoints_periodic_gru_sequence_Fw measured-versus-predicted TE curve

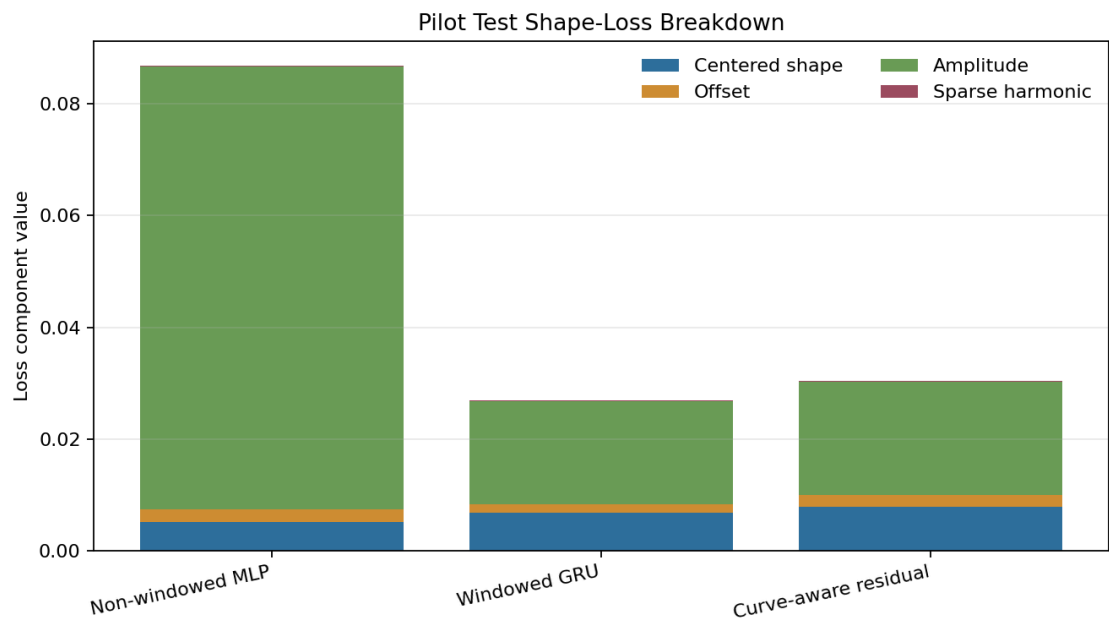


polished_setpoints_periodic_gru_sequence_Fw measured-versus-predicted TE curve

Secondary Scalar Diagnostics



Pilot scalar metric summary



Pilot shape loss breakdown

Technical Interpretation

The scalar outcome rejects further investment in the v3 windowed GRU branch as the immediate next candidate. It improved enough to remain informative, but the non-windowed MLP produced lower validation MAE, lower test MAE, lower test RMSE, and lower centered curve-shape loss in this pilot.

The curve-aware residual branch recovered from a poor first epoch but finished behind both leading neural candidates on scalar error. Its structured residual diagnostics remain useful for understanding offset behavior, but it should not be expanded before a stronger scalar or curve-screen signal appears.

The practical next step is a bounded shape-gated `TE Curve Verification Pipeline` screen for the non-windowed MLP winner, comparing it explicitly against the current windowed forward reference and the best non-windowed forward reference. Do not promote from this campaign leaderboard alone.

Closeout Decision

The campaign is closed as a successful pilot with no training failures. The recommended candidate for the next bounded curve-first screen is `shape_objective_periodic_mlp_harmonic_fw`. The windowed GRU continuation and curve-aware residual arms should remain as negative/secondary evidence unless the curve screen contradicts the scalar ranking.